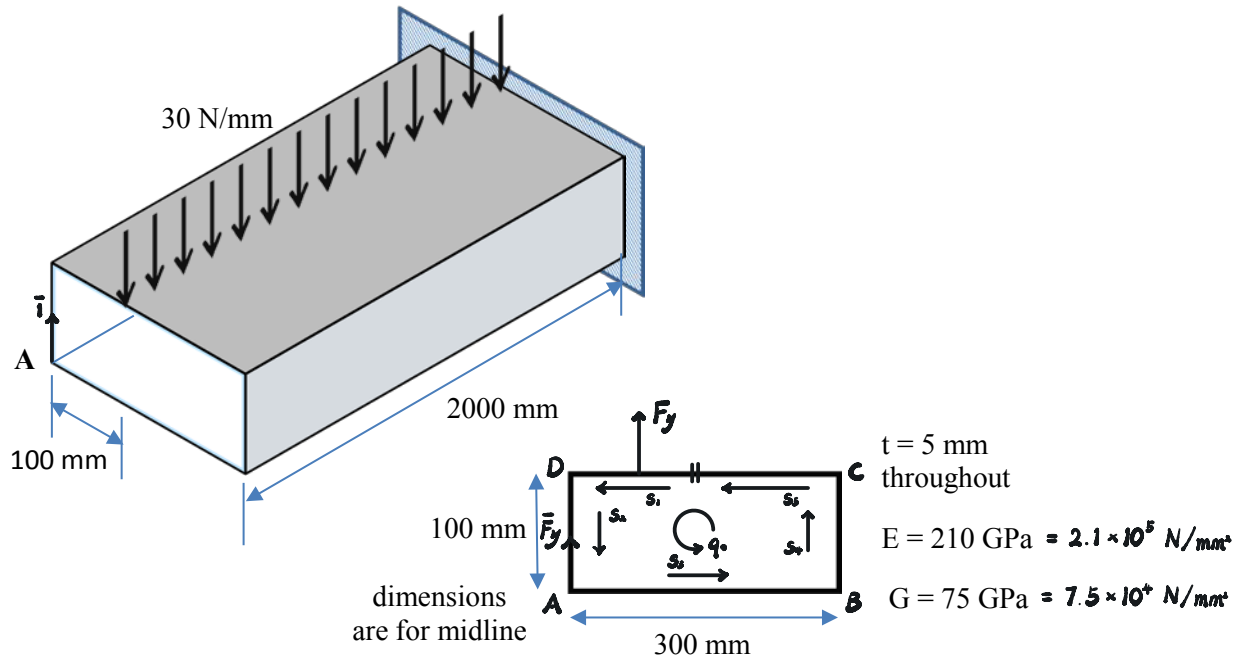


Imperial College London
Department of Aeronautics
Structures 3 - Tutorial Sheet 3

1. The thin walled, single cell beam of rectangular cross section and constant thickness is shown in the figure. The beam is built-in at one end and subjected to a uniformly distributed vertical loading of 30 N/mm along a line 100mm from the left side. Determine the vertical displacement at corner A at the free end of the beam.



$$I_x = 2 \cdot \frac{5 \cdot 100^3}{12} + \frac{300 \cdot 5^3}{12} + (50 - 0)^2 \cdot (300 \cdot 5) + \frac{300 \cdot 5^3}{12} + (0 - 50)^2 \cdot (300 \cdot 5)$$

$$= 8.33 \times 10^6 \text{ (mm}^4\text{)}$$

cut the close-cell at the middle of DC, $q = \frac{F_y}{I_x} D_x$

$$(D_x)_1 = - \int_0^{s_1} 50 \cdot 5 \, ds_1 = -250 s_1 \quad \therefore \int q_1 \cdot ds_1 = \frac{F_y}{I_x} (-125 s_1^2)$$

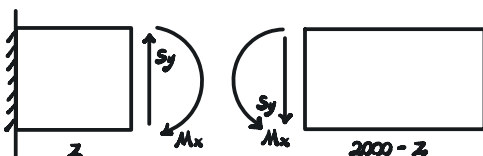
$$(D_x)_2 = -37500 - \int_0^{s_2} (50 - s_2) \cdot 5 \, ds_2 = \frac{5}{2} s_2^2 - 250 s_2 - 37500 \quad \therefore \int q_2 \cdot ds_2 = \frac{F_y}{I_x} \left(\frac{5}{6} s_2^3 - 125 s_2^2 - 37500 s_2 \right)$$

$$(D_x)_3 = -37500 - \int_0^{s_3} -50 \cdot 5 \, ds_3 = 250 s_3 - 37500 \quad \therefore \int q_3 \cdot ds_3 = \frac{F_y}{I_x} (125 s_3^2 - 37500 s_3)$$

$$(D_x)_4 = 37500 - \int_0^{s_4} (s_4 - 50) \cdot 5 \, ds_4 = -\frac{5}{2} s_4^2 + 250 s_4 + 37500 \quad \therefore \int q_4 \cdot ds_4 = \frac{F_y}{I_x} \left(-\frac{5}{6} s_4^3 + 125 s_4^2 + 37500 s_4 \right)$$

$$(D_x)_5 = 37500 - \int_0^{s_5} 50 \cdot 5 \, ds_5 = -250 s_5 + 37500 \quad \therefore \int q_5 \cdot ds_5 = \frac{F_y}{I_x} (-125 s_5^2 + 37500 s_5)$$

for real work,



$$\sum F = S_y + 30(2000 - z) = 0 \quad \therefore S_y = 30(z - 2000)$$

$$\sum M = M_x - \int_z^{2000} 30(2000 - z) \, dz = 0$$

$$\therefore M_x = 15(2000 - z)^2 \quad \therefore F_y = 30(z - 2000)$$

for virtual work, $\bar{F}_y = \bar{I}$, $\bar{m}_x = -\bar{I} \cdot (2000 - z) = z - 2000$

$$\therefore v_b = \int_0^{2000} \frac{15(2000-z)^2(z-2000)}{EI_x} dz = -\frac{15}{EI_x} \int_0^{2000} (2000-z)^3 dz$$

$$= \frac{15}{4EI_x} [(2000-z)^4]_0^{2000} = -34.299 \text{ (mm)}$$

when applying F_y , take moment about C.

$$-200 \cdot F_y = 2q_0(300 \cdot 100) + 100 \cdot \int_0^{300} q_3 ds_3 + 300 \cdot \int_0^{100} q_2 ds_2$$

$$= 60000 q_0 + 0 - 150 F_y$$

$$\therefore q_0 = \left(\frac{7}{1200} F_y \right) - \frac{1}{1200} F_y$$

when applying $\bar{F}_y$, take moment about C.

$$-300 \cdot \bar{F}_y = 60000 \bar{q}_0 + 0 - 150 \bar{F}_y$$

$$\therefore \bar{q}_0 = \left[\frac{3}{400} \bar{F}_y = \frac{3}{400} \right] - \frac{1}{400}$$

$$v_s = \int_0^1 \oint \frac{q \bar{q}}{tG} ds dz = \int_0^1 \oint \frac{q_3 \bar{q}_3}{tG} ds dz + \int_0^1 \oint \frac{q_2 \bar{q}_2}{tG} ds dz$$

$$\text{where } \int_0^1 \oint \frac{q_3 \bar{q}_3}{tG} ds dz$$

$$= \frac{30}{tG I_x^2} \int_0^{2000} (z-2000) \left[\int_0^{100} (-250s_1)^2 ds_1 + \int_0^{100} \left(\frac{5}{2} s_2^2 - 250s_2 - 37500 \right)^2 ds_2 + \int_0^{300} (250s_3 - 37500)^2 ds_3 \right. \\ \left. + \int_0^{100} \left(-\frac{5}{2} s_4^2 + 250s_4 + 37500 \right)^2 ds_4 + \int_0^{100} (-250s_5 + 37500)^2 ds_5 \right] dz$$

$$= \frac{30}{tG I_x^2} \cdot (7.03125 \times 10^{10} + 1.7396 \times 10^{10} + 1.40625 \times 10^{10} + 1.7396 \times 10^{10} + 7.03125 \times 10^{10}) \cdot \int_0^{2000} z - 2000 dz$$

$$= \frac{30}{tG I_x^2} \cdot 6.2917 \times 10^{10} \cdot (-2 \times 10^6) = -1.451 \text{ (mm)}$$

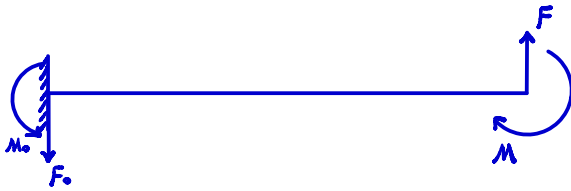
$$\text{and } \int_0^1 \oint \frac{q_2 \bar{q}_2}{tG} ds dz$$

$$= \frac{1}{tG} \cdot \left(\frac{7}{1200} \cdot \frac{3}{400} \right) \cdot 30 \int_0^{2000} (z-2000) \cdot \oint ds dz$$

$$= \frac{\textcircled{21} 1}{16000 tG} \cdot (2 \cdot 300 + 2 \cdot 100) \cdot \int_0^{2000} z - 2000 dz = \textcircled{-5.6} \text{ (mm)}$$

$$\therefore v = v_b + v_s = -34.299 - (1.451 + \textcircled{5.6}) = \textcircled{-41.35} \text{ (mm)}$$

OR



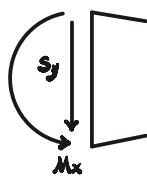
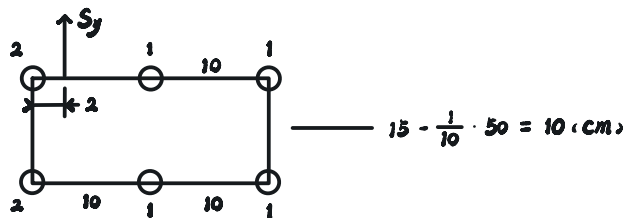
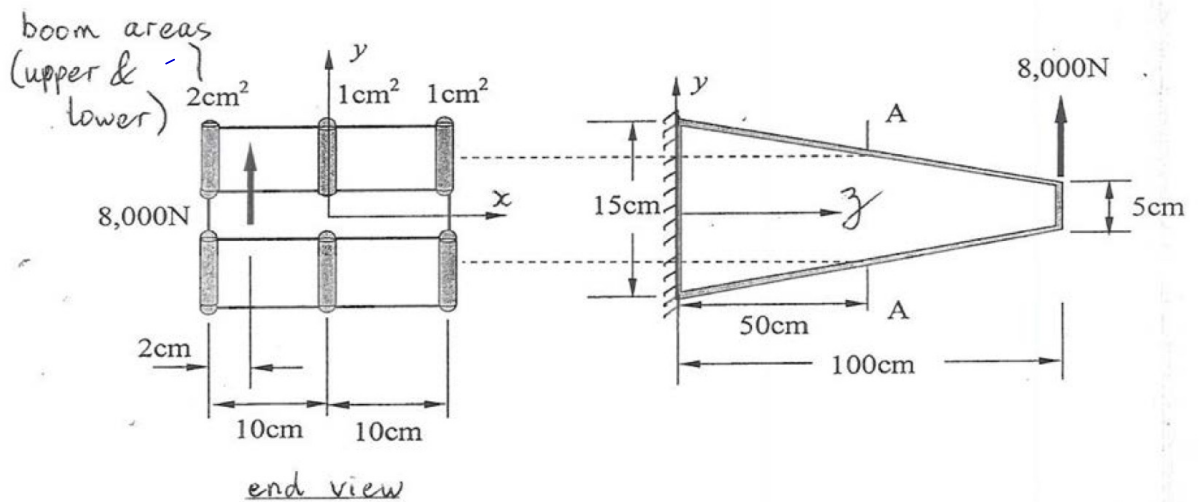
$$\Sigma F = 0 \longrightarrow -F_0 - 30 \cdot 2000 = 0 \quad \therefore F_0 = -60000$$

$$\Sigma M = 0 \longrightarrow -M_0 + 15 \cdot 2000^2 = 0 \quad \therefore M_0 = 15 \cdot 2000^2$$

$$\therefore \Sigma F = 0 \longrightarrow -F_0 - 30z + F = 0 \quad \therefore F = 30z - 60000$$

$$\Sigma M = 0 \longrightarrow -M_0 - F_0 z - 15z^2 + M = 0 \quad \therefore M = 15(2000^2 - 4000z + z^2) = 15(2000 - z)^2$$

2. The figure below shows a boom shear panel idealisation of a symmetrically tapered cantilever beam. Find the shear flow distribution at section A-A.



$$S_y = F = 8000 \text{ (N)}$$

$$M_x = -F \cdot (L - z) = -8000 \cdot (100 - 50) = -4 \times 10^5 \text{ (N} \cdot \text{cm)} = -4000 \text{ (Nm)}$$

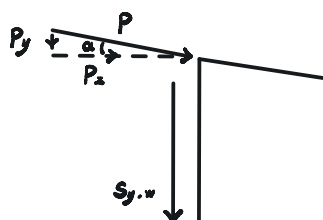
$$I_x = (5 - 0)^2 (2 + 1 + 1) + (0 - 5)^2 (2 + 1 + 1) = 200 \text{ (cm}^4\text{)} = 2 \times 10^{-6} \text{ (m}^4\text{)}$$

$$\therefore (s_x)_{top} = \frac{M_x}{I_x} \cdot \frac{5}{100} = -1 \times 10^8 \text{ (Pa)}$$

$$\therefore (s_x)_{bottom} = 1 \times 10^8 \text{ (Pa)}$$

$$\therefore (P_x)_1 = (s_x) \cdot A_1 = \pm 1 \times 10^8 \cdot \frac{1}{100^2} = \mp 10000 \text{ (N)}$$

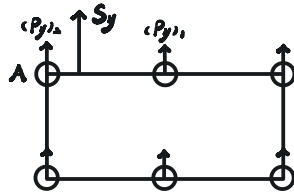
$$(P_x)_2 = (s_x) \cdot A_2 = \pm 1 \times 10^8 \cdot \frac{2}{100^2} = \mp 20000 \text{ (N)}$$



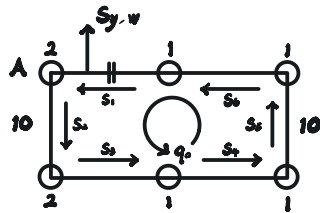
$$P_y = P_x \tan \alpha = P_x \frac{5}{100} = \frac{P_x}{20}$$

$$\therefore (P_y)_1 = 500 \text{ (N)}$$

$$(P_y)_2 = 1000 \text{ (N)}$$



$$\therefore S_{y,w} = S_y - 2(P_y)_2 - 4(P_y)_1 = 4000 \text{ (N)} \quad \therefore q = \frac{S_{y,w}}{I_x} D_x = 2 \times 10^4 D_x$$



$$(D_x)_{s1} = -\int_0^L y t \, ds_1 = 0, \quad (D_x)_{s2} = -(5-0) \cdot 2 = -10 \text{ (cm}^2\text{)} = -1 \times 10^{-5} \text{ (m}^2\text{)}$$

$$(D_x)_{s3} = -10 - (-10) = 0, \quad (D_x)_{s4} = 0 - (-5) = 5 \times 10^{-6} \text{ (m}^2\text{)}$$

$$(D_x)_{s5} = 5 - (-5) = 1 \times 10^{-5} \text{ (m}^2\text{)}, \quad (D_x)_{s6} = 10 - 5 = 5 \times 10^{-6} \text{ (m}^2\text{)}$$

$$\therefore q_1 = 0, \quad q_2 = -2 \times 10^4 \text{ (N/m)}, \quad q_3 = 0, \quad q_4 = 1 \times 10^4 \text{ (N/m)}, \quad q_5 = 2 \times 10^4 \text{ (N/m)}, \quad q_6 = 1 \times 10^4 \text{ (N/m)}$$

Take moment about A.

$$S_y \cdot \frac{2}{100} = 2q_0 \left(\frac{20}{100} \cdot \frac{10}{100} \right) + \left[\frac{10}{100} (0 + 1 \times 10^4) \cdot \frac{10}{100} + \frac{20}{100} \cdot 2 \times 10^4 \cdot \frac{10}{100} \right] \\ + \left[2(P_y)_1 \cdot \frac{10}{100} + 2(P_y)_2 \cdot \frac{20}{100} \right]$$

$$\therefore \frac{1}{50} S_y = \frac{1}{25} q_0 + (100 + 400) + (100 + 200)$$

$$\therefore q_0 = 25(160 - 800) = -16000 \text{ (N/m)}$$

$\therefore q = q_0 + q_b$ as shown below

